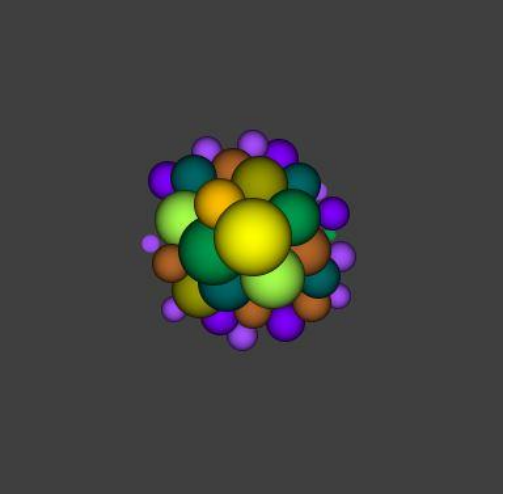
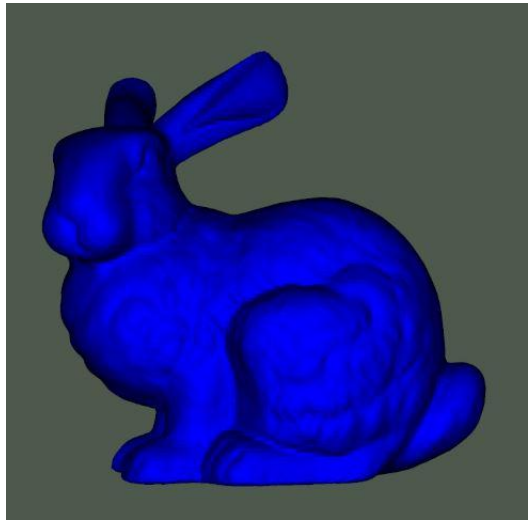


CS 440: Computer Graphics

Week 4

Lecture 7: Affine Geometry



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

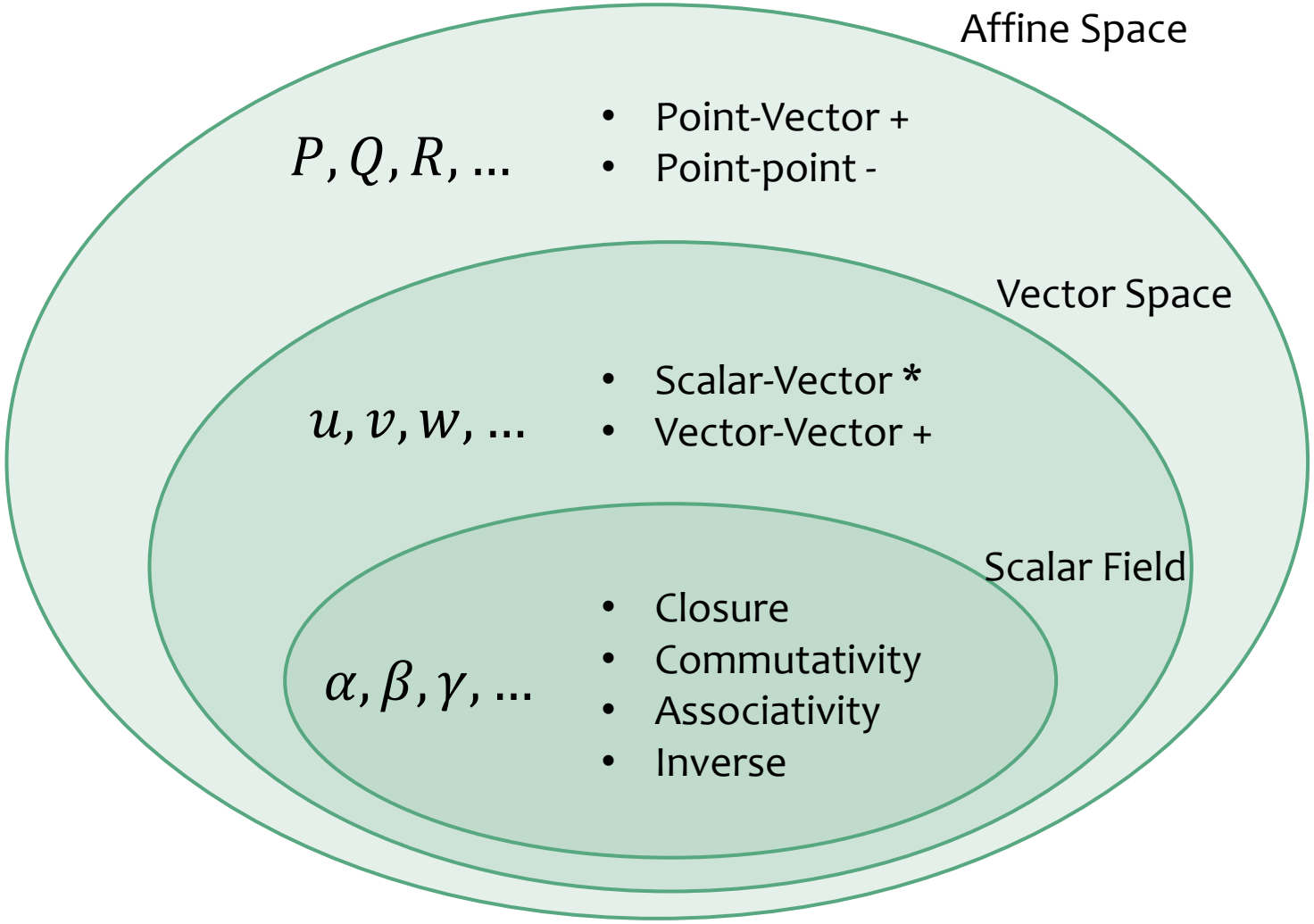
- Assignment 2 released
  - Feel free to post your Sierpinskis and Mandelbrots on the discussion thread if you think they look cool!
- Quiz during class on Wednesday
- Recitations Feedback?

- Recap:
  - Graphics Pipeline
  - WebGL Functions
  - Affine Space

Today: Affine Geometry

- Affine Space
- Affine Geometry: Geometric Primitives and Operations
- 3D Primitives
- Coordinate System
- Change in Coordinate System
- Coordinate Frame
- Change in Coordinate Frame
- Homogenous Coordinates

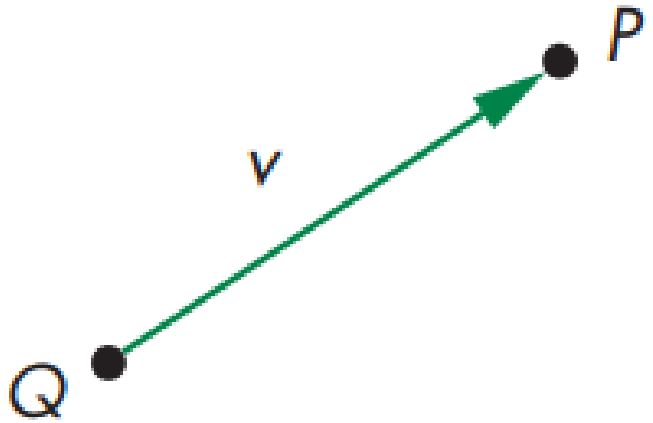
Recall: Affine Space



Recall: Point-Vector Operations

- Points can be added with vectors to displace a point

$$P = Q + v$$



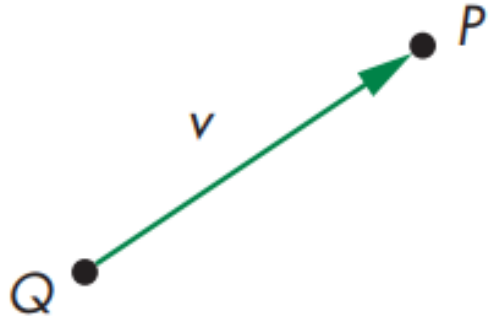
- Point-point subtraction yields a vector

$$v = P - Q$$

Geometric Primitives: Line

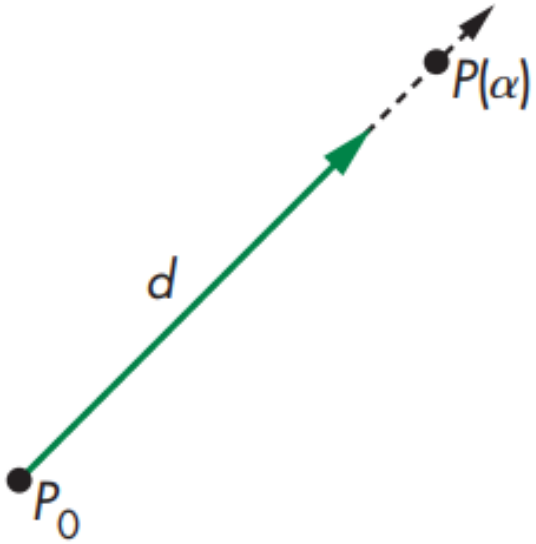
- Previously: Points can be added with vectors to displace a point

$$P = Q + v$$



- Parametric form of a line
- Sum of point and vector is a **line**

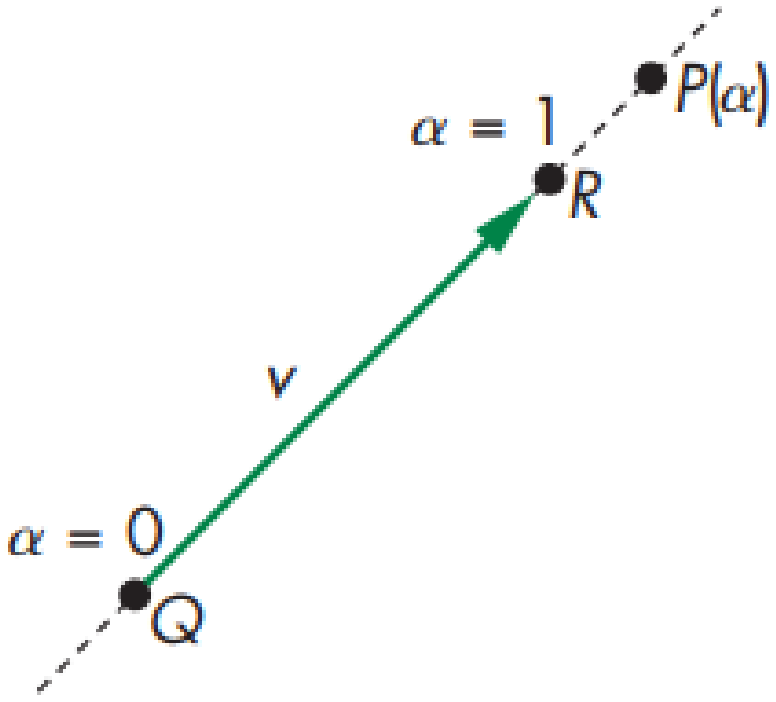
$$P(\alpha) = P_0 + \alpha d$$



Affine Sum

- Point-point addition does not exist
- Affine sum is always equivalent to 1

$$P = \alpha R + (1 - \alpha)Q$$

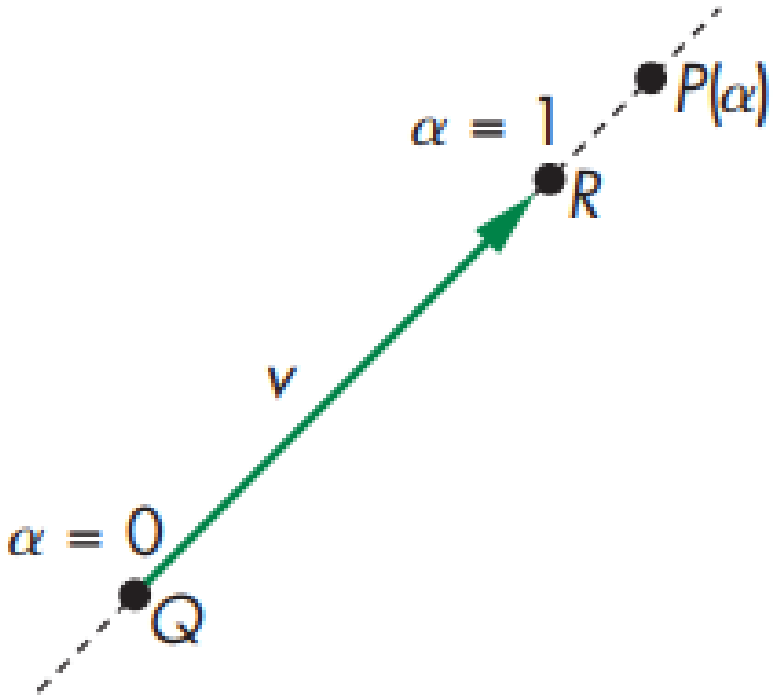


Affine Sum

$$P(\alpha) = Q + \alpha v$$

$$P(\alpha) = Q + \alpha(R - Q)$$

$$P(\alpha) = \alpha R + (1 - \alpha)Q$$



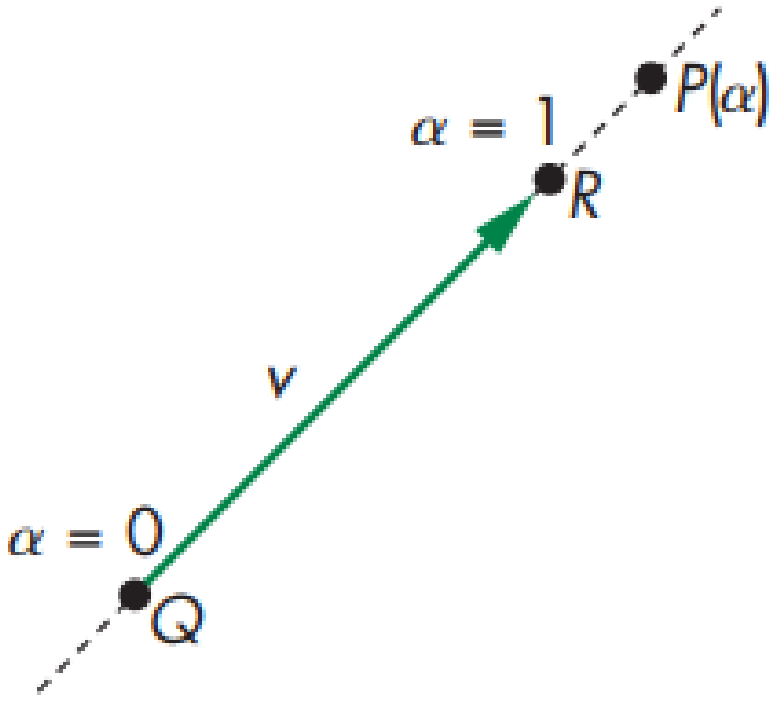
Affine Sum

$$P(\alpha) = \alpha R + (1 - \alpha)Q$$

$$P(\alpha) = \alpha_1 R + \alpha_2 Q$$

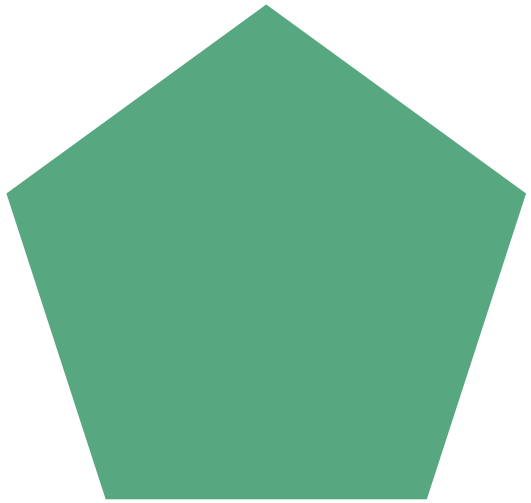
such that

$$\alpha_1 + \alpha_2 = 1$$



Convexity

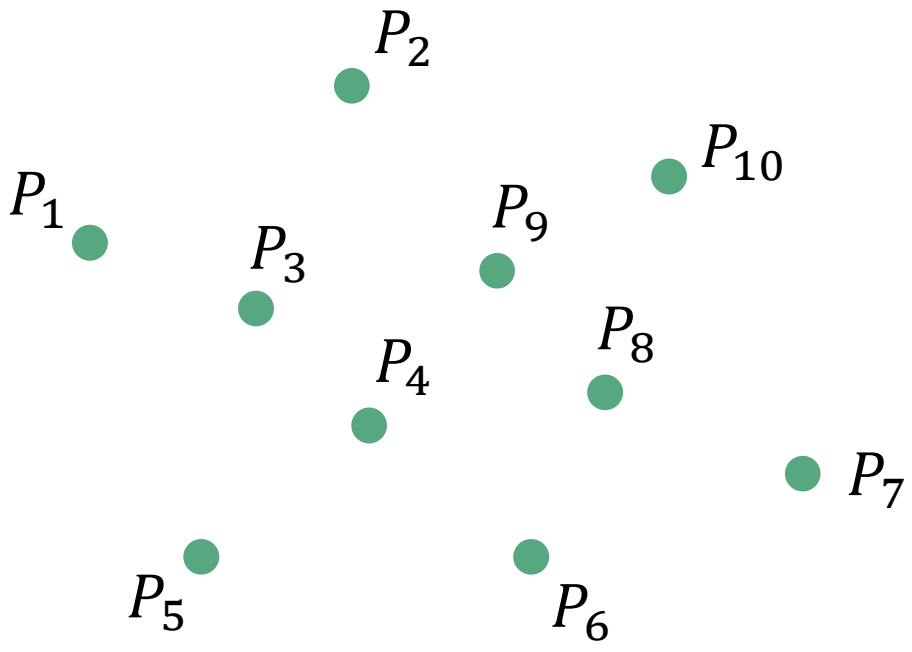
- **Convex polygon:** Any line segment between two points of the polygon stays entirely within or on the polygon.



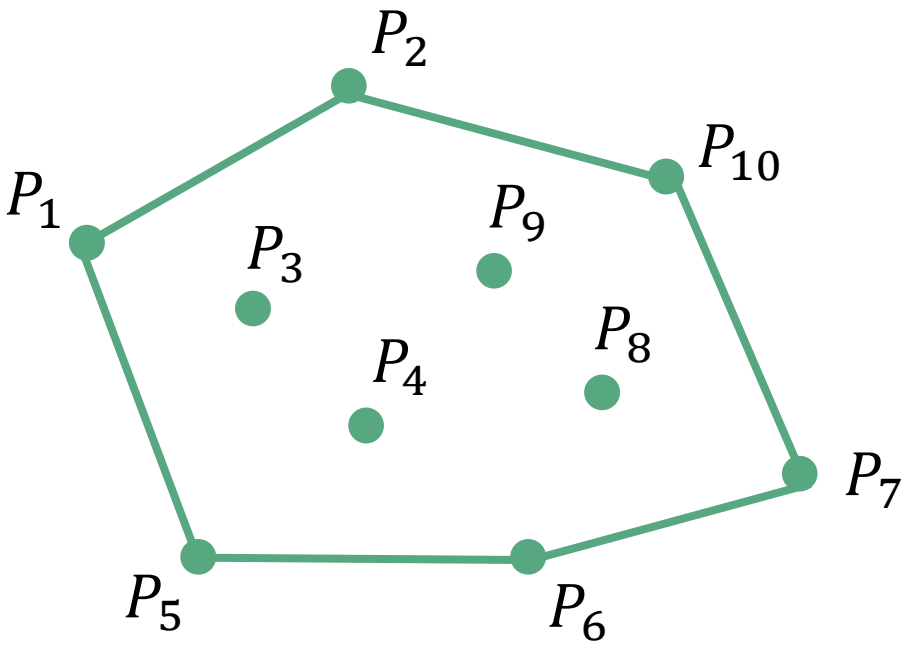
Convex

Non Convex

Convex Hull



Convex Hull



Convex Hull and Affine Sum

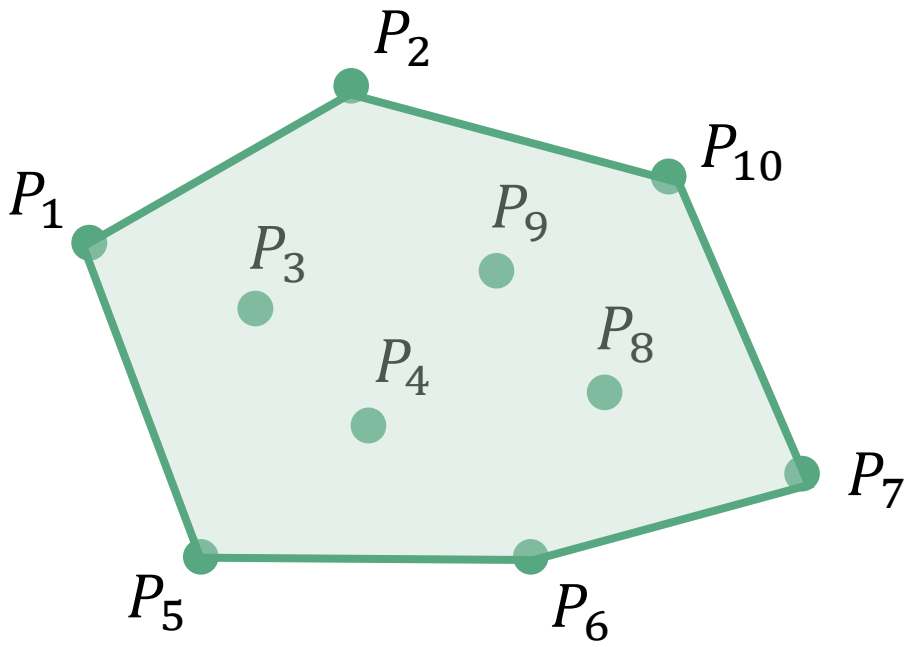
$$P = \alpha_1 P_1 + \alpha_2 P_2 + \ldots + \alpha_{10} P_{10}$$

Affine sum:

$$\alpha_1 + \alpha_2 + \ldots + \alpha_{10} = 1$$

Convexity:

$$\alpha_1 \geq 0, \alpha_2 \geq 0, \ldots, \alpha_{10} \geq 0$$

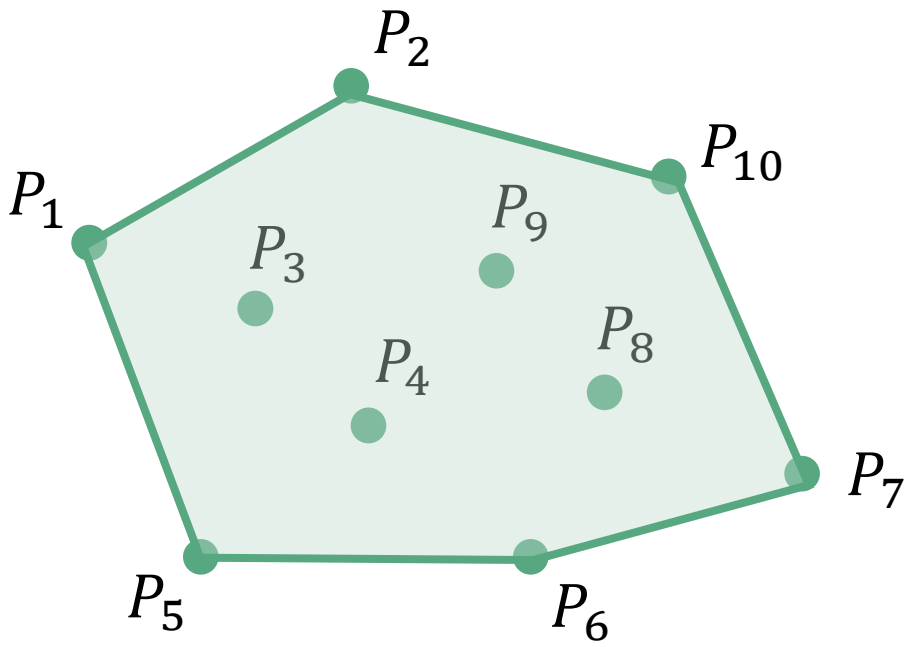


Convex Hull and Affine Sum

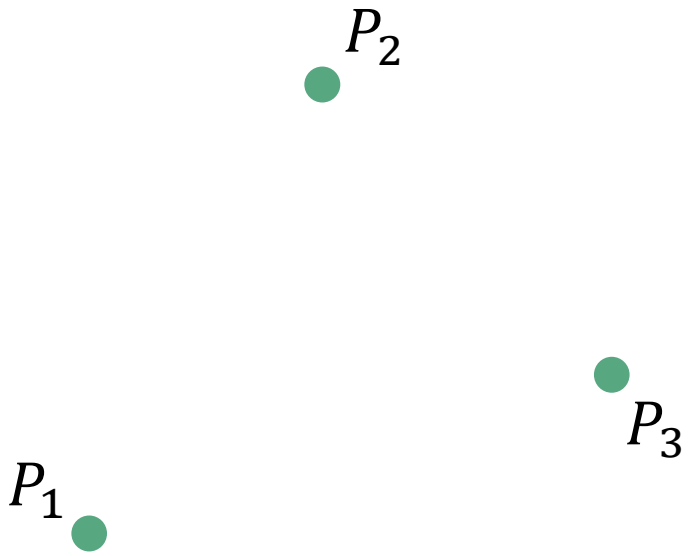
$$P = \sum_i \alpha_i P_i$$

$$\sum_i \alpha_i = 1$$

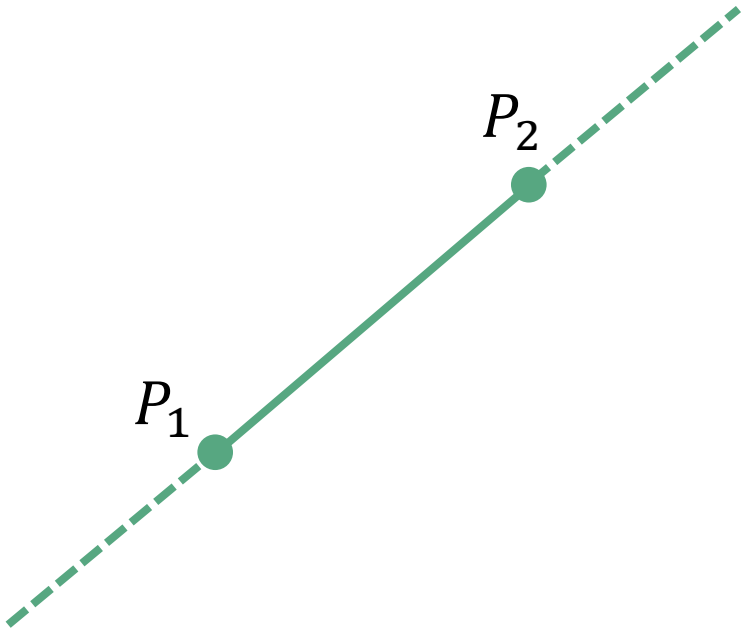
$$\forall i \; \alpha_i \geq 0$$

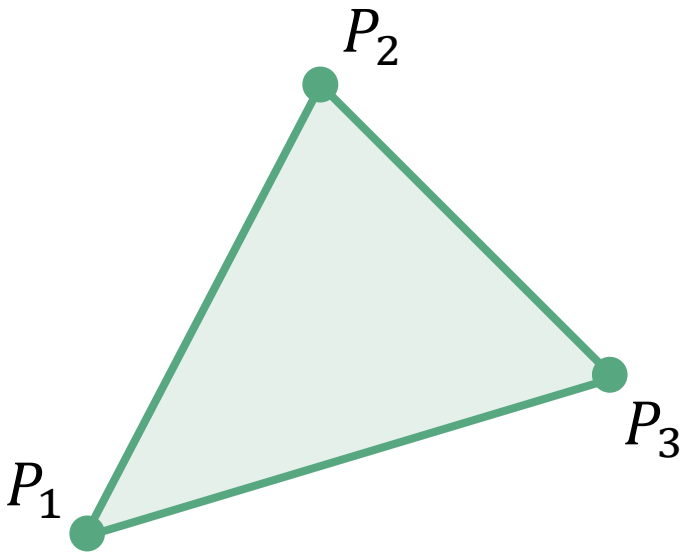


Convexity of points?



Convex Sum: Line and Triangle





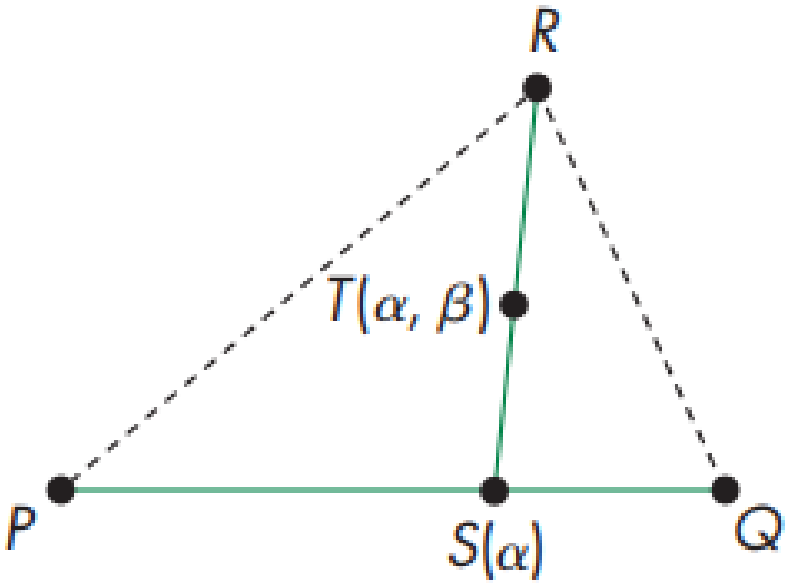
Geometric Primitives: Plane

- Three non-collinear points form a plane

$$S(\alpha) = \alpha P + (1 - \alpha)Q, \qquad 0 \leq \alpha \leq 1$$

$$T(\beta) = \beta S + (1 - \beta)R, \qquad 0 \leq \beta \leq 1$$

$$T(\alpha, \beta) = \beta[\alpha P + (1 - \alpha)Q] + (1 - \beta)R$$



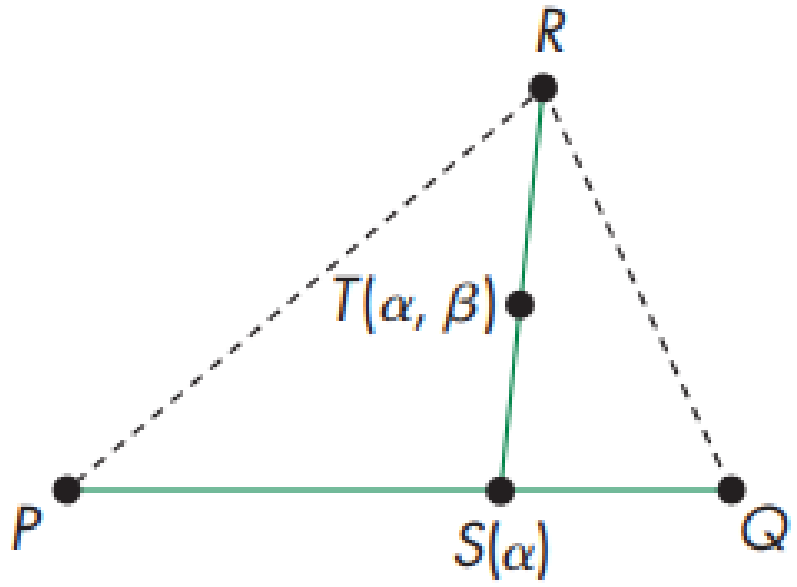
Geometric Primitives: Plane

- Equation of plane in vector form

$$T(\alpha,\beta) = \alpha\beta P + \beta(1-\alpha)Q + (1-\beta)R$$

$$T(\alpha,\beta) = P + \alpha(Q - P) + \beta(R - P)$$

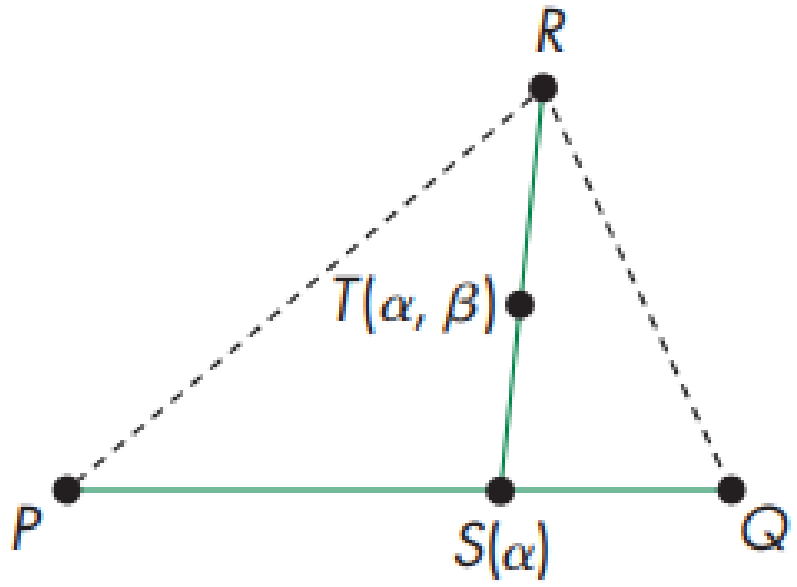
$$T(\alpha,\beta) = P_0 + \alpha u + \beta v$$



Barycentric Coordinates

- Equation of plane in vector form

$$T(\alpha, \beta) = \alpha\beta P + \beta(1-\alpha)Q + (1-\beta)R$$



$$T(\alpha, \beta) = \alpha' P + \beta' Q + \gamma' R$$

such that

$$\alpha' + \beta' + \gamma' = 1$$

Barycentric Coordinates: Triangle

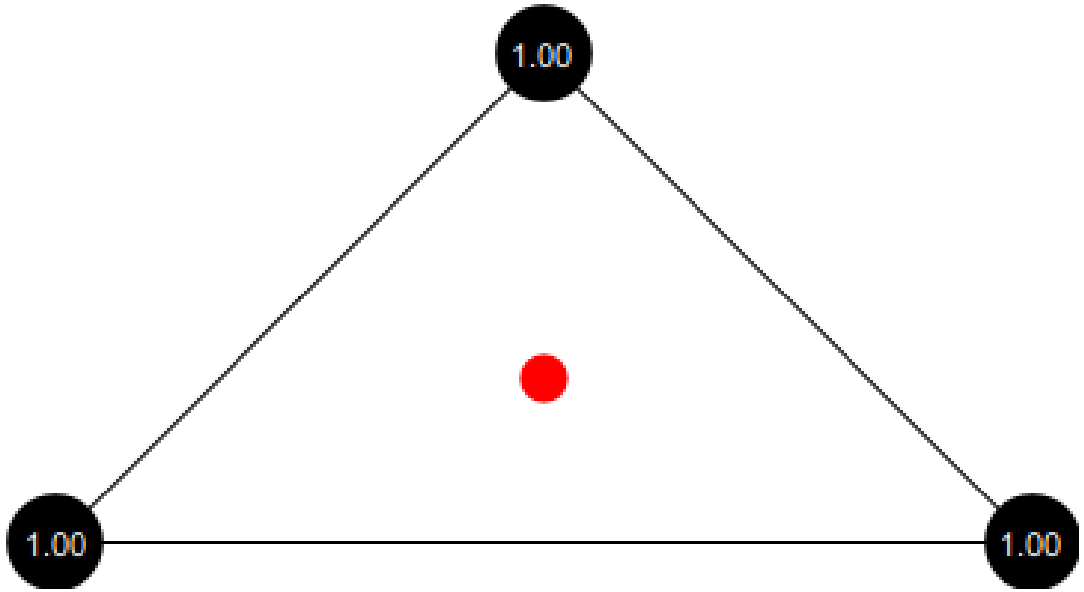
$$T(\alpha, \beta) = \alpha' P + \beta' Q + \gamma' R$$

such that

$$\alpha' + \beta' + \gamma' = 1$$

- For any point T inside a triangle,

$$0 < \alpha' < 1$$
$$0 < \beta' < 1$$
$$0 < \gamma' < 1$$



Why Affine Spaces?

- https://youtu.be/Yyvk84QFwwI?si=31HDTng_aKwaW2f2&t=292

Normals

$$P = P_0 + \alpha u + \beta v$$

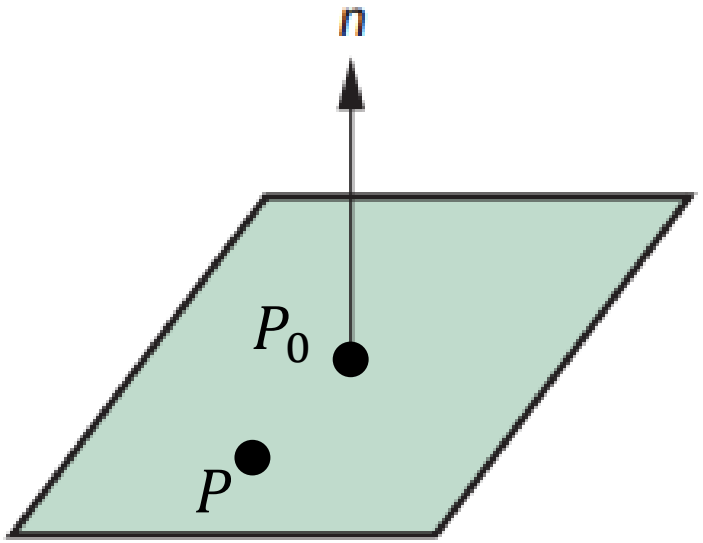
$$P - P_0 = \alpha u + \beta v$$

- Vector that is orthogonal to both u and v

$$n = u \times v$$

$$n \cdot (P - P_0) = (\alpha u) \cdot n + (\beta v) \cdot n$$

$$n \cdot (P - P_0) = 0$$



Putting to perspective: Normals

- https://youtu.be/L3oCQEwtMLI?si=HPajTJzTBA6M48ee